

Introduction

First, I'm gonna introduce some of the key terminologies before moving on with the main topic of today's presentation. Let's start with Neural Networks. These are powerful models that learn patterns from structured data, like images, which are grids of pixels, or text, which is a sequence of words. They work really well when the data has a clear, regular structure.

But traditional NNs have a big limitation—they struggle with irregular or non-grid data. And many real-world systems are not grids at all, they're graphs.

A graph is simply a collection of nodes, representing entities, and edges, representing relationships. Think of people and friendships in a social network, or atoms and chemical bonds in a molecule. Graphs are everywhere. In fact, almost any data can be represented by graphs. For example, in an image, pixels can be denoted as nodes and edges can represent the relationship between pixels. Similarly, a sentence can be visualized as a graph where words are nodes and edges represent relationships between adjacent words.

This is where Graph Neural Networks, or GNNs, come in. GNNs they enable machine learning tasks that leverage the interconnected nature of data. They can learn from both the features of the nodes and the connections between them

This image is an example of a complex real world data which has to be represented as a graph and is commonly used to represent social networks like Facebook where the nodes are users and the edges between them are friendships.

OVERVIEW

GNNs are deep learning models built for graphs that can capture complex relationships and dependencies. Unlike traditional models, they learn directly from raw graph structures, making them powerful tools for graph-based data.

So in short, GNNs allow us to bring the power of neural networks to graph-structured data

TASKS of GNN

*Graph learning tasks usually fall into three main categories. First, **node-level prediction**, where we focus on individual nodes—like detecting whether a user in a financial network is fraudulent. Second, **edge-level prediction**, A classic use case is a friendship recommendation for two people who have similar social media connections or recommending products. And third, **graph-level prediction**, where we look at the entire graph—like determining whether a molecule is toxic or could work as a drug, for example in the discovery of the antibiotic Halicin.*

Significance :

- **Handling Non-Euclidean Data:** Traditional neural networks (NNs), such as CNNs (for grids/images) and RNNs (for sequences/text), struggle with data structured as graphs because graphs possess **non-Euclidean structures** that lack a fixed shape or predefined ordering.
- **Arbitrary Input Shapes:** GNNs are designed to function even when the size and shape of the input graph changes arbitrarily within a dataset
- **Permutation Invariance:** GNNs produce the same results regardless of how the nodes in the graph are ordered.
- Rich Feature Extraction:** GNNs learn **rich representations (embeddings)** directly and optimally from the raw graph data
- **Leveraging Network Context:** Predictions made using GNNs leverage the **surrounding graph context**, meaning predictions on connected nodes and edges inform each other
- **Inductive Capability:** Certain GNN variants, such as Graph Attention Networks (GATs) and GraphSAGE, are suitable for **inductive learning**, allowing the model to work with graph structures or new nodes that it has never encountered during training.
- Drug Discovery and Molecular Design:** GNNs accelerate the development of medicines and materials. They were instrumental in the discovery of the antibiotic **Halicin**
- **Simulations and Engineering:** GNNs can **shorten the time required for complex simulations**,
- **Personalized Healthcare:** They are useful in developing models to **predict chemotherapy response** for individual cancer patients.
- **Anomaly and Fraud Detection:** GNNs model is effective for tasks like detecting suspicious transactions in financial networks or identifying anomalous edges in large enterprise security systems.
- **Scalable Recommendation Systems:** GNNs are widely used in modern platforms (like Pinterest and Uber Eats) to improve personalized recommendations by capturing complex relationships efficiently.

GNN Architecture :

*Graph Neural Networks operate through message passing in four main steps. First, **initialization**: each node starts with a feature vector that describes its basic properties. Next comes **message passing**, where each node gathers information from its neighbors, often by summing or averaging their features. Then, during the **update** step, the node combines this new information with its existing features to create an updated representation. Finally, **stacking layers** allows this process to repeat across multiple rounds, letting each node learn from neighbors of neighbors and capture broader patterns across the graph. In the end, every node ends up with a refined embedding that reflects both its own properties and its position within the larger network.*

Graph Neural Networks are like students learning a concept together. Each student starts with their own understanding, then shares and gathers knowledge from classmates. By repeating this over several rounds, they learn from friends of friends too, building a deeper and more complete understanding of the concept

TYPES:

*There are several popular Graph Neural Network architectures, each differing in how they aggregate and update information. **Graph Convolutional Networks, or GCNs**, combine information from all a node's neighbors to update its features. **Graph Attention Networks, or GATs**, take this a step further by giving more weight to the most important neighbors, so the model focuses on what really matters.*

GraphSAGE stands for Graph SAmple and aggreGatE. which makes it scalable to very large graphs. GraphSAGE operates via a two-step process: Sampling and Aggregation.

1. Sampling (The "SA" in SAGE):

- Instead of processing a node's entire neighborhood, GraphSAGE employs a fixed-size neighbor sampling strategy.*

2. Aggregation (The "GE" in SAGE):

- After sampling a fixed set of neighbors, the model gathers and aggregates feature information from these sampled neighbors.*

- The model then concatenates the information of the target node (its own feature vector) with the aggregated information from its neighbors.*

GraphSAGE approaches large graphs efficiently by sampling a fixed number of neighbors and combining their features, allowing it to scale to very big networks without losing important information."

USE case : SNA

Graph Neural Networks play a key role in social network analysis. They can **detect communities** and classify users, **recommend friends or content** through link prediction, and help platforms like Airbnb understand host behavior for **trust and safety**, even with limited data. GNNs are also used for **anomaly and abuse detection**, such as spotting rumors or fake news by analyzing propagation and user patterns. Scalability is handled with models like GraphSAGE, which sample neighbors to process massive networks efficiently, while **dynamic GNNs** track evolving connections in real-time as social networks grow and change

PINTEREST

*Pinterest is a visual discovery platform where users save and organize pins into boards. Its massive network—2 billion pins, 1 billion boards, and over 18 billion connections—is called the Pinterest graph. To make better recommendations, Pinterest is currently actively deploying **PinSage**, a GNN-powered recommendation system scaled-up to operate on the Pinterest graph that predicts which pins users might like based on their interests and connections. Compared to traditional content-based methods, PinSage achieves **150% higher hit-rate** and **60% higher mean reciprocal rank over the best baseline model**, making recommendations far more accurate and relevant.*

USE CASE : MB

Graph Neural Networks are transforming science and healthcare by modeling complex relationships. In **drug discovery and material science**, GNNs predict molecular properties and interactions, speeding up the identification of promising candidates, such as the antibiotic Halicin. They also help predict **protein-protein interactions**, analyze **genomic sequences**, and personalize cancer treatments by combining tumor images with gene expression data. Beyond medicine, GNNs are used to reduce **methane emissions** in agriculture and study **brain connectivity**, capturing dynamic patterns in neural networks for insights into cognition and neurological disorders.

PROTEIN DESIGN

Protein design aims to create new proteins with specific properties, which can be used in drugs, enzymes, or new materials. Traditional methods are slow and costly, but Baker Lab has developed **RFDiffusion**, an AI system combining Graph Neural Networks and diffusion models. Using an E(n)-Equivariant GNN, it can design protein structures that meet custom constraints. Since its release in 2022, RFDiffusion has dramatically outperformed previous methods—solving nearly all benchmark problems and achieving up to **214 times higher experimental success** for certain proteins. Experts see it as one of the biggest breakthroughs in structural biology in recent years, alongside AlphaFold

CHALLENGES

Training GNNs on large graphs is challenging due to high memory and computational demands. Storing the full graph and node features can quickly exceed memory limits for massive networks, and updating node embeddings recursively leads to long training times. A major issue is the **neighborhood expansion problem**: as layers are stacked, each node needs information from neighbors of neighbors, causing exponential growth in computation. Additionally, graphs are irregular, unordered, and sometimes dynamic, which adds complexity compared to fixed structures like images or text

SAMPLING STRATEGIES

To make GNNs scalable for large graphs, researchers use various sampling techniques to reduce computational cost while maintaining performance.

Node-wise sampling selects neighbors for each node—GraphSAGE, for example, samples a fixed number of neighbors to accelerate training and handle unseen nodes.

Layer-wise sampling chooses nodes per layer to limit neighborhood growth.

Graph-wise sampling extracts entire subgraphs, enabling efficient mini-batch training and better memory usage. These strategies are essential for large-scale applications, like recommender systems or urban data analytics, where full-graph training is impractical.

FUTURE DIRECTIONS

Future GNN research is focused on making models more scalable and efficient. This involves techniques like reducing edges while keeping important information, graph coarsening, and multi-resolution approaches. Advances in GPU and TPU computing, as well as distributed learning, help handle large datasets. Researchers are also tackling challenges like sampling trade-offs and designing models for complex graphs, including heterogeneous and dynamic networks.

GNN IN AI RESEARCH

Graph Neural Networks have seen rapid growth in research. Looking at recent ICLR and NeurIPS, which are two major annual conferences which focus on cutting edge AI research, GNNs consistently appear as one of the top three keywords, highlighting their increasing importance in cutting-edge AI research.